Gadha Krishna Leons

New Delhi, India
+91-9886149085 | gadha.leons@gmail.com
gadhak.github.io/gadha_portfolio | gadhak

Professional Summary

Final-year Ph.D. researcher in Cancer Genomics specializing in the molecular characterization of B-cell Acute Lymphoblastic Leukemia (B-ALL). Proven track record of integrating high-throughput transcriptomics (RNA-Seq) with clinical metadata to identify cryptic subtypes and predictive biomarkers in BCR-ABL1-negative cohorts. Expertise in developing custom bioinformatics pipelines (R, Python, WGCNA) and validating in-silico findings through rigorous wet-lab assays (Sanger, MLPA, qPCR). Aiming to translate large-scale genomic data into actionable biological insights by combining bioinformatics pipelines with molecular biology in a postdoctoral role.

Education

- **Ph.D. in Cancer Genomics** — *2022 – 2026*
  All India Institute of Medical Sciences (AIIMS) — New Delhi, India
  - Study of predictive and prognostic molecular genetic aberrations in BCR::ABL1 negative B-ALL.
- **Advanced PG Diploma in Life Science Technologies** — *2020 – 2021*
  SRM-DBT Platform, SRMIST — Chennai, India
- **M.Sc. in Applied Genetics** — *2017 – 2019*
  Bangalore University — Bangalore, India
  - CGPA: 7.8 (1st Class Distinction)
- **B.Sc. (Genetics, Microbiology, Biochemistry)** — *2014 – 2017*
  Ramaiah College of Arts, Science and Commerce — Bangalore, India
  - CGPA: 8.42 (1st Class Exemplary)

Research Experience

- **Junior Research Fellow / Ph.D. Scholar** — *Oct 2021 - Present*
  AIIMS, New Delhi — New Delhi, India
  - Led comprehensive transcriptomic profiling of B-ALL samples to identify high-risk subtypes (Ph-like) and novel fusions.
  - Engineered and optimized bioinformatics pipelines using **R** and **Python** to analyze RNA-Seq data (**DGE analysis**, **WGCNA)**.
  - Validated in-silico findings using wet-lab techniques including **Sanger sequencing**, **RT-qPCR**, and **MLPA**.
  - Discovered prognostic biomarkers by correlating transcriptomic signatures with clinical survival data.
- **PG Diploma Researcher** — *Jan 2021 - July 2021*
  SRM-DBT Platform, SRMIST — Chennai, India
  - Conducted phylogenetic analysis and clustering of SARS-CoV-2 genomes circulating in the South Indian population.
  - Performed molecular docking studies to model protein-protein interactions and analyze spike protein variants.
- **Master's Dissertation Researcher** — *Jan 2019 - July 2019*
  Bangalore University — Bangalore, India
  - Characterized sperm morphological diversity in Caecilians (Amphibia: Gymnophiona) of Western Ghats.

TECHNICAL SKILLS

- **Bioinformatics**
  - **NGS Analysis:** Bulk RNA-Seq Analysis (STAR, DESeq2), Single-Cell RNA-Seq Analysis (Seurat)
  - **Methods & Concepts:** Weighted Gene Coexpression Network Analysis (WGCNA), Pathway Enrichment (GSEA, ClusterProfiler, Enrichr, ssGSEA),
  - **Structural:** Protein Structure Modeling, Protein-Protein Docking
- **Programming**
  - R (Bioconductor, Tidyverse, ggplot2), Python, Linux/Unix, Git/Version Control
- **Molecular Biology (Wet Lab)**
  - NGS Library Prep (Illumina), Sanger Sequencing, MLPA, RT-qPCR, Flow Cytometry, DNA/RNA Extraction
- **General Research:** Cell Culture, Zebrafish Genetic Engineering (CRISPR, Microinjection)

PUBLICATIONS

[1] **Krishna Leons, G**., Gupta, S. K., Sharma, P., et al. (In Review). **Decoding the 'B-Other' Black Box: Integrated RNA-Sequencing and Machine Learning Unveils Cryptic B-ALL Subtypes in a Single Centre Study.**

[2] **Krishna Leons, G**., Sharma, P., Gupta, S. K., et al. (2026). **Concurrent Retinoblastoma 1 (RB1) and IKZF1 Deletions Have Adverse Outcome in Childhood B-Cell Acute Lymphoblastic Leukemia (B-ALL).** *Int J Lab Hematol*. DOI:10.1111/ijlh.70069

[3] Gupta, S. K., **Leons, G. K.**, Sharma, P., et al. (2025). **Application of Targeted RNA-Sequencing in High-Risk B-Cell Acute Lymphoblastic Leukemia (B-ALL).** *Int J Lab Hematol*. DOI: 10.1111/ijlh.14551

[4] Sharma, P., **Leons, G. K.**, Singh, V., et al. (2025). **Gene expression signature of IKZF1 deleted B-cell acute lymphoblastic leukaemia reveals a unique regulation of MUC4.** *Br J Haematol*. DOI: 10.1111/bjh.70014

[5] Gupta, S.K., **Leons, G. K.** (2025). **Genetics of B-Cell Acute Lymphoblastic Leukemia (B-ALL): Recent Updates and Indian Perspective.** *Indian J Hematol Blood Transfus*. DOI: 10.1007/s12288-025-01988-y

[6] **Leons, G. K.**, Sharma, P., Gupta, S. K., et al. (2023). **Detection of a novel SNX2 gene breakpoint in the SNX2::ABL1 fusion transcript in Ph-like B-cell acute lymphoblastic leukemia.** *Pediatric Blood & Cancer*. DOI: 10.1002/pbc.30546

[7] Sharma, P., Gupta, S. K., **Leons, G. K.,** et al. (2022). **IKZF1::CIITA, a novel fusion transcript in a case of relapsed B-cell acute lymphoblastic leukemia**. *Pediatric blood & cancer*, 70(1), e29872. DOI: 10.1002/pbc.29872

[8] Gupta, S. K., & **Leons, G. K.** (2025). **Molecular Diagnostics.** In Sharma SK (Ed.), *Manual of Hematology* (pp. 158-169). Jaypee Brothers Medical Publishers.

[9] Gupta, S. K., & **Leons, G. K.** (2025). **Diagnosis of Ph-like B-ALL: The Ideal Approach in the Indian Setting**. In *Recent Advances in Hematology-5* (pp. 92-103). Jaypee Brothers Medical Publishers.

CONFERENCE PRESENTATIONS

- **Poster Presentation** *2026*
  2026 Korean Society of Hematology (KSH) International Conference & 67th Annual Meeting Seoul, Korea
  The Seeds of Relapse: Uncovering the Dormant Transcriptomic Signatures of Future Relapse at Diagnosis

- **Oral Presentation** *2025*
  87th Annual Meeting of the Japanese Society of Hematology (JSH) Kobe, Japan
  Transcriptomic Signatures and Fusion Landscape in Relapsed B-ALL: Insights from RNA-Sequencing
- **Poster Presentation** *2025*
  Molecular Analysis for Precision Oncology Congress (ESMO-MAP) Paris, France
  Transcriptomic Profiling of CDKN2A/B-Deleted B-ALL Reveals Key Coding and Non-Coding Biomarkers
- **Oral Presentation** *2024*
  2024 Korean Society of Hematology (KSH) International Conference & 65th Annual Meeting Seoul, Korea
  Adverse Clinical Impact of RB1 Gene Deletions Alone and with Concurrent IKZF1 Gene Deletions in Pediatric B-ALL
- **Abstract Publication** *2025*
  European Hematology Association (EHA)
  Whole Transcriptome Sequencing Uncovers Key Coding and Non-Coding Biomarkers in IKZF1Plus B-ALL.

HONORS AND AWARDS

- **ICKSH Travel Award** *2026*
  2026 Korean Society of Hematology (KSH) International Conference & 67th Annual Meeting (ICKSH)
- **JSH Travel Award** *2025*
  87th Annual Meeting of the Japanese Society of Hematology (JSH)
- **ICMR International Travel Grant** *2025*
  Molecular Analysis for Precision Oncology (MAP) Congress-European Society for Medical Oncology (ESMO)
- **ANRF International Travel Grant** *2024*
  2024 Korean Society of Hematology (KSH) International Conference & 65th Annual Meeting (ICKSH)
- **JGEEBILS Qualified** *2021*
  Tata Institute of Fundamental Research (TIFR)
  - Reference No.: GS2021BIOPHD030224
- **GATE Biotechnology (AIR 1289)** *2020*
  Indian Institute of Technology (IIT)
  - Scored 475 in Graduate Aptitude Test in Engineering.

CERTIFICATIONS

- **NASA GeneLab | GL4U: RNA Sequencing Module Set** *Aug 2025*
- **NASA GeneLab | GL4U: Introduction Module Set** *Jul 2025*
- **Training in "Engineering Zebrafish with CRISPR Tools"** (CSIR-CCMB) *Mar 2020*
- **Certificate in Animal Cell Culture Techniques** (CellKraft Biotech) *Feb 2020*
- **Certificate in Genetic Counseling** (IMAeVarsity& Martin Luther Christian University)

REFERENCES

Dr. Sanjeev Kumar Gupta
Professor (PhD Supervisor)
AIIMS, New Delhi
drskgupta1@gmail.com

Dr. Ritu Gupta
Professor
AIIMS, New Delhi
drritugupta@gmail.com